

We first suppose $A_N = \omega(1)$, further breaking into cases which result in different terms inside D and E becoming dominant:

$$\text{If } A_N = \Omega(N), \quad D = \Theta \left(A_N^{\frac{3}{4}} N^{\frac{3}{4}} (\log N)^{\frac{3}{8}} \log A_N \right), \quad E = \Theta \left(A_N N \sqrt{\log(N) \log(A_N)} \right).$$

$$\text{If } A_N = \Omega \left(\frac{N^{\frac{1}{3}}}{\sqrt{\log N}} \right), A_N = o(N), \quad D = \Theta \left(A_N^{\frac{3}{4}} N^{\frac{3}{4}} (\log N)^{\frac{3}{8}} \log N \right), \quad E = \Theta(A_N N \log N).$$

$$\text{If } A_N = \omega(1), A_N = o \left(\frac{N^{\frac{1}{3}}}{\sqrt{\log N}} \right), \quad D = \Theta(N \log N), \quad E = \Theta(A_N N \log N).$$

In each case, $E = \omega(D)$ and so $D - E \rightarrow -\infty$, contradicting that $D = E$. Thus a stationary point requires $A_N = \mathcal{O}(1)$ for a stationary point.

We now do the same for $A_N = o(1)$. First, clearly $n \geq 1$; suppose that in fact $n = \Theta(1)$, i.e. $A_N = \Theta(1/(N\sqrt{\log N}))$. In this case, we would have $D = \frac{2}{3}N \log \frac{n}{\delta} + \Theta(1)$ and $E = N + \Theta(1)$, so that $D = E$ requires $\frac{2}{3} \log \frac{n}{\delta} \rightarrow 1$, i.e. $n \rightarrow \delta \exp \frac{3}{2} \approx 4.5 \delta$. For $\delta < 0.22$, this contradicts $n \geq 1$. So we know that $\log n = \omega(1)$. Now, the remaining options for A_N all yield $D - E \rightarrow \infty$:

$$\text{If } A_N = o(1), A_N = \Omega \left(\frac{1}{\log N} \right), \quad D = \Theta(N \log n), \quad E = \Theta(A_N N \log n).$$

$$\text{If } A_N = o \left(\frac{1}{\log N} \right), A_N = \omega \left(\frac{1}{N\sqrt{\log N}} \right), \quad D = \Theta(N \log n), \quad E = \Theta(N).$$

Thus we have established that $A_N = \Theta(1)$. Thus, we obtain that

$$D = \frac{1}{2}N \log N + \mathcal{O}(N) \quad E = \frac{\sqrt{3}J^*}{2C} A_N N \log N + \mathcal{O}(N \sqrt{\log N}).$$

Asymptotic equality hence requires $A_N \rightarrow C/(\sqrt{3}J^*)$. □

A.3. Uniform convergence results

These results, on the uniform convergence of $\hat{\eta}$ and $\hat{\sigma}^2$, were used in the proof of Theorem 11.

Proposition 15. *Under Assumptions (A) to (C), we have that with probability at least $1 - \delta$,*

$$\sup_{\omega} |\hat{\eta}_{\omega} - \eta_{\omega}| \leq \frac{8}{\sqrt{n}} \left[\nu \sqrt{2 \log \frac{2}{\delta} + 2D \log(4R_{\Omega} \sqrt{n})} + L_k \right].$$

Proof. Theorem 7 of [Sriperumbudur et al. \(2009\)](#) gives a similar bound in terms of Rademacher chaos complexity, but for ease of combination with our bound on convergence of the variance estimator, we use a simple ϵ -net argument instead.

We study the random error function

$$\Delta(\omega) := \hat{\eta}_{\omega} - \eta_{\omega}.$$

First, we place T points $\{\omega_i\}_{i=1}^T$ such that for any point $\omega \in \Omega$, $\min_i \|\omega - \omega_i\| \leq q$; Assumption (B) ensures this is possible with at most $T = (4R_{\Omega}/q)^D$ points ([Cucker & Smale, 2001](#), Proposition 5).

Now, $\mathbb{E} \Delta = 0$, because $\hat{\eta}$ is unbiased. Recall that $\hat{\eta} = \frac{1}{n(n-1)} \sum_{i \neq j} H_{ij}$, and via Assumption (A) we know $|H_{ij}| \leq 4\nu$. This $\hat{\eta}$, and hence Δ , satisfies bounded differences: if we replace (X_1, Y_1) with (X'_1, Y'_1) , obtaining $\hat{\eta}' = \frac{1}{n(n-1)} \sum_{i \neq j} F_{ij}$ where F agrees with H except when i or j is 1, then

$$\begin{aligned} |\hat{\eta} - \hat{\eta}'| &\leq \frac{1}{n(n-1)} \sum_{i \neq j} |H_{ij} - F_{ij}| = \frac{1}{n(n-1)} \sum_{i>1} |H_{i1} - F_{i1}| + \frac{1}{n(n-1)} \sum_{j>1} |H_{1j} - F_{1j}| \\ &\leq \frac{2}{n(n-1)} \sum_{i>1} 8\nu = \frac{16\nu}{n}. \end{aligned}$$

Using McDiarmid's inequality for each $\Delta(\omega_i)$ and a union bound, we then obtain that with probability at least $1 - \delta$,

$$\max_{i \in \{1, \dots, T\}} |\Delta(\omega_i)| \leq \frac{16\nu}{\sqrt{2n}} \sqrt{\log \frac{2T}{\delta}} \leq \frac{8\nu}{\sqrt{n}} \sqrt{2 \log \frac{2}{\delta} + 2D \log \frac{4R_\Omega}{q}}.$$

We also have via Assumption (C), for any two $\omega, \omega' \in \Omega$,

$$\begin{aligned} |\hat{\eta}_\omega - \hat{\eta}_{\omega'}| &\leq \frac{1}{n(n-1)} \sum_{i \neq j} |H_{ij}^{(\omega)} - H_{ij}^{(\omega')}| \leq \frac{1}{n(n-1)} \sum_{i \neq j} 4L_k \|\omega - \omega'\| = 4L_k \|\omega - \omega'\| \\ |\eta_\omega - \eta_{\omega'}| &= |\mathbb{E}[H_{12}^{(\omega)}] - \mathbb{E}[H_{12}^{(\omega')}]| \leq \mathbb{E}|H_{12}^{(\omega)} - H_{12}^{(\omega')}| \leq 4L_k \|\omega - \omega'\| \end{aligned}$$

so that $\|\Delta\|_L \leq 8L_k$. Combining these two results, we know that with probability at least $1 - \delta$

$$\sup_{\omega} |\Delta(\omega)| \leq \max_{i \in \{1, \dots, T\}} |\Delta(\omega_i)| + 8L_k q \leq \frac{8\nu}{\sqrt{n}} \sqrt{2 \log \frac{2}{\delta} + 2D \log \frac{4R_\Omega}{q}} + 8L_k q;$$

setting $q = 1/\sqrt{n}$ yields the desired result. $\square$

Proposition 16. Under Assumptions (A) to (C), with probability at least $1 - \delta$,

$$\sup_{\omega \in \Omega} |\hat{\sigma}_\omega^2 - \sigma_\omega^2| \leq \frac{64}{\sqrt{n}} \left[7 \sqrt{2 \log \frac{2}{\delta} + 2D \log (4R_\Omega \sqrt{n})} + \frac{18\nu^2}{\sqrt{n}} + 8L_k \nu \right].$$

Proof. We again use an ϵ -net argument on the (random) error function

$$\Delta(\omega) := \hat{\sigma}_{k_\omega}^2 - \sigma_{k_\omega}^2.$$

First, choose T points $\{\omega_i\}_{i=1}^T$ such that for any point $\omega \in \Omega$, $\min_i \|\omega - \omega_i\| \leq q$; again, via Assumption (B) and Proposition 5 of Cucker & Smale (2001) we have $T \leq (4R_\Omega/q)^D$. By Lemmas 17 and 18 and a union bound, with probability at least $1 - \delta$,

$$\max_{i \in \{1, \dots, T\}} |\Delta(\omega_i)| \leq 448 \sqrt{\frac{2}{n} \log \frac{2T}{\delta}} + \frac{1152\nu^2}{n} \leq 448 \sqrt{\frac{2}{n} \log \frac{2}{\delta} + \frac{2}{n} D \log \frac{4R_\Omega}{q}} + \frac{1152\nu^2}{n}.$$

Lemma 19 shows that $\|\Delta\|_L \leq 512L_k\nu$, which means that with probability at least $1 - \delta$,

$$\sup_{\omega \in \Omega} |\Delta(\omega)| \leq 448 \sqrt{\frac{2}{n} \log \frac{2}{\delta} + \frac{2}{n} D \log \frac{4R_\Omega}{q}} + \frac{1152\nu^2}{n} + 512L_k\nu q. \quad (9)$$

Taking $q = 1/\sqrt{n}$ gives the desired result. $\square$

Lemma 17. For any kernel k bounded by ν (Assumption (A)), with probability at least $1 - \delta$,

$$|\hat{\sigma}_k^2 - \mathbb{E} \hat{\sigma}_k^2| \leq 448 \sqrt{\frac{2}{n} \log \frac{2}{\delta}}.$$

Proof. We simply apply McDiarmid's inequality to $\hat{\sigma}_k^2$. Suppose we change (X_1, Y_1) to (X'_1, Y'_1) , giving a new H matrix F which agrees with H on all but the first row and column. Note that $|H_{ij}| \leq 4\nu$, and recall

$$\hat{\sigma}_k^2 = 4 \left(\frac{1}{n^3} \sum_i \left(\sum_j H_{ij} \right)^2 - \left(\frac{1}{n^2} \sum_{ij} H_{ij} \right)^2 \right).$$

The first term in the parentheses of $\hat{\sigma}_k^2$ changes by

$$\left| \frac{1}{n^3} \sum_i \left(\sum_j H_{ij} \right)^2 - \frac{1}{n^3} \sum_i \left(\sum_j F_{ij} \right)^2 \right| \leq \frac{1}{n^3} \sum_{ij\ell} |H_{ij}H_{i\ell} - F_{ij}F_{i\ell}|.$$

In this sum, if none of i, j , or ℓ are one, the term is zero. The n^2 terms for which $i = 1$ are each upper-bounded by $32\nu^2$, simply bounding each H or F by 4ν . Of the remainder, there are $(n-1)$ terms where $j = \ell = 1$, each $|H_{i1}^2 - F_{i1}^2| \leq 16\nu^2$. We are left with $2(n-1)^2$ terms which have exactly one of j or ℓ equal to 1; the $j = 1$ terms are $|H_{i1}H_{i\ell} - F_{i1}H_{i\ell}| \leq |H_{i1} - F_{i1}||H_{i\ell}| \leq (8\nu)(4\nu)$, so each of these terms is at most $32\nu^2$. The total sum is thus at most

$$\frac{1}{n^3} (n^2 32\nu^2 + (n-1)16\nu^2 + 2(n-1)^2 32\nu^2) = \left(\frac{6}{n} - \frac{7}{n^2} + \frac{3}{n^3} \right) 16\nu^2.$$

The remainder of the change in $\hat{\sigma}_k^2$ can be determined by bounding

$$\begin{aligned} \left| \sum_{ij} H_{ij} - \sum_{ij} F_{ij} \right| &\leq \sum_{ij} |H_{ij} - F_{ij}| = \sum_j |H_{1j} - F_{1j}| + \sum_{i>1} |H_{i1} - F_{i1}| \\ &\leq n(8\nu) + (n-1)(8\nu) = (8\nu)(2n-1), \end{aligned}$$

which then gives us

$$\begin{aligned} \left| \left(\frac{1}{n^2} \sum_{ij} H_{ij} \right)^2 - \left(\frac{1}{n^2} \sum_{ij} F_{ij} \right)^2 \right| &= \left| \frac{1}{n^2} \sum_{ij} H_{ij} + \frac{1}{n^2} \sum_{ij} F_{ij} \right| \left| \frac{1}{n^2} \sum_{ij} H_{ij} - \frac{1}{n^2} \sum_{ij} F_{ij} \right| \\ &\leq (2 \cdot 4\nu) \frac{2n-1}{n^2} (8\nu) = 64\nu^2 \left(\frac{2}{n} - \frac{1}{n^2} \right). \end{aligned}$$

Thus

$$|\hat{\sigma}_k^2 - (\hat{\sigma}'_k)^2| \leq 4 \left[\left(\frac{6}{n} - \frac{7}{n^2} + \frac{3}{n^3} \right) 16\nu^2 + \left(\frac{2}{n} - \frac{1}{n^2} \right) 64\nu^2 \right] = \frac{64\nu^2}{n^3} [14n^2 - 11n + 3] \leq \frac{896\nu^2}{n}.$$

Because the same holds for changing any of the (X_i, Y_i) pairs, the result follows by McDiarmid's inequality. $\square$

Lemma 18. *For any kernel k bounded by ν (Assumption (A)), the estimator $\hat{\sigma}_k^2$ satisfies*

$$|\mathbb{E} \hat{\sigma}_k^2 - \sigma_k^2| \leq \frac{1152\nu^2}{n}.$$

Proof. We have that

$$\mathbb{E} \hat{\sigma}_k^2 = 4 \left(\frac{1}{n^3} \sum_{ij\ell} \mathbb{E} [H_{i\ell} H_{j\ell}] - \frac{1}{n^4} \sum_{ijab} \mathbb{E} [H_{ij} H_{ab}] \right).$$

Most terms in these sums have their indices distinct; these are the ones that we care about. (We could evaluate the expectations of the other terms exactly, but it would be tedious.) We can thus break down the first term as

$$\begin{aligned} \frac{1}{n^3} \sum_{ij\ell} \mathbb{E} [H_{i\ell} H_{j\ell}] &= \frac{1}{n^3} \sum_{ij\ell: |\{i,j,\ell\}|=3} \mathbb{E} [H_{i\ell} H_{j\ell}] + \frac{1}{n^3} \sum_{ij\ell: |\{i,j,\ell\}|<3} \mathbb{E} [H_{i\ell} H_{j\ell}] \\ &= \frac{n(n-1)(n-2)}{n^3} \mathbb{E} [H_{12} H_{13}] + \left(1 - \frac{n(n-1)(n-2)}{n^3} \right) q, \end{aligned}$$